Linus Behrbohm

Systems & Data Visualization Engineer

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LinkedIn | GitHub



Software engineer for systems and interactive data visualization with Rust, TypeScript, and WebAssembly. Builds understandable views of real data plus graph viewers and editors for examining agent and model workflows. Available remotely or on site in Hamburg.

Languages: German, English
Programming Languages: Rust, TypeScript, Python, Go, Bash, C++
Web / UI: Vue.js, Node.js, Vite, WebAssembly, Dioxus, egui, Qt
AI / Agents: Rig, TensorFlow, Jupyter Notebook
Platform / Tools: Azure, Kubernetes, Docker, Git, Linux, Windows, LaTeX

Projects

Context Graph Engine

2020 – Present [repo]

Experimental language-model framework Rust, symbolic graph attention

- **Symbolic Language Model** – Built an original experimental language model and framework in Rust with symbolic graph attention, based on my bachelor's thesis.
- **Interpretable Representation** – Implemented reconstructable, continuously adaptable language and context representations in Rust, with graph viewers and editors for examining model behaviour.

Agent Builder

2026 – Present [repo]

Agent harnesses Rust, Rig

- **Agent Harnesses** – Developed own agent harnesses with Rig.

Workflow Tools

2025 – Present [repo]

Tools for resumable agent sessions Rust

- **Resumable Agent Workflows** – Built Rust tools that preserve context and task state across sessions and support validation and long-term tracking; individual developers at Nidec tested and used the tools.

Experience

Nidec SSB Wind Systems GmbH

03/2024 – 10/2026 **2.5+ years**

UI Product Owner & Platform Engineer Full-time Hybrid

- **HMI and LCT** – Further development of the HMI charging interface (approximately 50% of the final product) with Vue.js, TypeScript, and PHP; Added features and fixed defects and stability issues; 30% higher session success (according to Nidec) and approximately 80% fewer customer complaints (personal estimate); Reported 50% fewer HMI bugs per quarter and 90% lower UI latency; Further developed the LCT service interface for configuring embedded charging systems (approximately 20% of the final product) with Vue.js, TypeScript, and PHP; Added features and improved stability and defect resolution; firmware-update process estimated to be 80% faster (Vue.js, TypeScript, PHP).
- **Embedded advertising player (Ads)** – Fully developed and delivered a time-synchronized, cloud-configurable video advertising player for embedded systems; Synchronized and configured playlist and branding data for embedded systems (TypeScript, Python, PHP).
- **Agent tooling** – Built Rust tools over 6–8 months for automated translation updates, npm license checks, ticket systems, and task planning and code-change specification; Individual developers tested and used the tools; presented AI-assisted development cycles internally and advised a team of up to ten embedded developers; Workflow Tools originated from this work and continues to be developed (Rust).

- **Firmware build pipeline** – Integrated existing firmware build scripts with Azure Pipelines and the internal file server; Enabled authenticated full firmware builds, versioned artifacts, and developer documentation; increased iteration speed by an estimated 10–20x based on personal experience (YAML, Bash, Python, Azure).

Further Experience

RoederZeibig GmbH & Co. KG

05/2019 – 04/2020 **1 year**

Junior Full Stack Developer Part-time Remote

- **Application development** – Application bootstrapped; application features and SQL database communication for event planning implemented (Java, SQL).
- **Mobile-first frontend** – Mobile-first frontend for event-planning software developed (Bootstrap, JavaScript).
- **Microservices** – Microservices for event-planning software developed (Kubernetes, Go, Rust).

Froglogic GmbH

02/2018 – 09/2018 **8 months**

Software Engineer — Internship Part-time On-site

- **Free-text validation** – Features for parsing and validating free text developed (C++, Qt).
- **Gherkin integration** – Gherkin integrated into test-authoring tools; support for Cucumber tests extended (Gherkin, Cucumber).

Education

B. Sc. Cognitive Informatics — Grade: 2.3

10/2018 – 03/2024

University of Bielefeld [Bachelor's thesis]

- **Applied Machine Learning** – Implemented a reinforcement-learning training system and scoring model in a Jupyter Notebook with TensorFlow to generate humorous titles from research abstracts.
- **Language Modelling** – Developed an algorithm for unsupervised learning from relationships between substrings; formally analysed mathematical properties and worst-case complexity.
- **Distributed Systems** – Implemented resilient matrix synchronization and quorum-based cluster management, addressing message ordering, timing, and network failures.
- **Human-Centered Accessibility** – Developed a browser-inspection application that identifies accessibility barriers against established standards and suggests improvements.

Software Tester — Foundation Level

2018

iSQI Group

- **Testing Foundations** – Passed training in systematic testing, black-box methods, static analysis, security considerations, and practical risk-based quality models.

Abitur — Grade: 2.2 Anne-Frank-Schule Bargteheide 2017